

Setup and configure a LAN using a switch and Ethernet cables in your lab

EXP NO : 04

DATE : 03/08/2020

AIM :

Setup and configure a LAN using a switch and Ethernet cables in your lab, assign static IP addresses to all devices, Verify connectivity using the ping command and demonstrate resource / data sharing across the network.

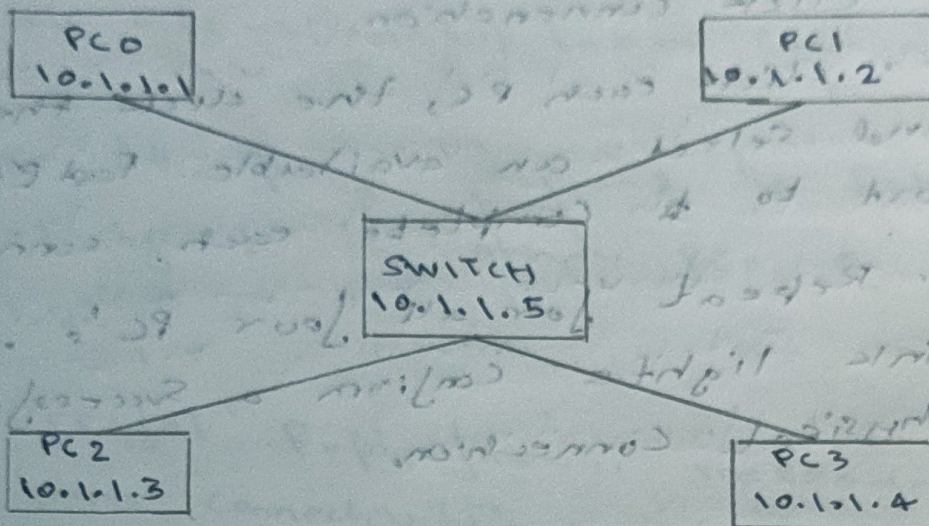
Equipment Required :

S. No	Equipment	Qty	Specification
1	Personal Computer (PC)	4	Ethernet (NIC) enabled
2	Network Switch	1	8/16/24 ports, unmanaged / managed
3	Ethernet cables (Cat5e / Cat6)	4	Straight-through RJ-45 connectors

Network Diagram :

The LAN was implemented using star topology as shown below, with 4 PCs connected individually to a central switch.

Star Topology with (4 PCs + Switch)



IP Address Configuration Table

Device	IP Address	Subnet Mask	Connected To
PC0	10.1.1.1	255.0.0.0	Switch Port Fa0/1
PC1	10.1.1.2	255.0.0.0	Switch Port Fa0/2
PC2	10.1.1.3	255.0.0.0	Switch Port Fa0/3
PC3	10.1.1.4	255.0.0.0	Switch Port Fa0/4
Switch (management)	10.1.1.5	255.0.0.0	VLAN 1 Interface

Implementation of Cisco PACKET

- Since physical hardware was limited, the LAN was also implemented and verified using Cisco PACKET TRACER simulation software following the steps below.

→ Step 1 :- Place the Devices

1. open Cisco Packet Tracer
2. From the device category panel
3. Select switch and switch 1.

→ Step 2: Cable the Device

- 4. click connection
- 5. click each PC, then click the switch and select an available Fast Ethernet port to complete each connection.
- 6. Repeat for all four PC's. Green link lights confirm a successful physical connection.

→ Step 3: Assign IP Address to PCs

- 7. Click a PC → Desktop tab → IP configuration
- 8. Enter the static IP address

→ Step 4: Configure the Switch

management IP (optional)

- 9. Click the Switch → CLI tab

- 10. Enter the following commands to assign management IP.

configure terminal
interface Vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit

→ Step 5 : Verify Connectivity using Ping

11. click a PC → Desktop tab → Command Prompt.

12. Use the ping Command to test connectivity to the other PCs, eg. Ping 10.1.1.2.

13. Successful "Reply from..." messages confirm connectivity.

→ Step 6 : Demonstrate Data Sharing (Server Simulation)

14. Add a Server device (End Devices → and connect to the switch. Server-RT)

15. Assign it an IP address via its Desktop → IP Configuration tab.

16. Open the Server Services (config) tab and enable the HTTP Service.

17. From another PC, open the Web Browser (Desktop tab) and enter the Server's IP address to load the page, confirming successful data transfer across the LAN.

→ Step 7 : observe Packet Flow in Simulation mode :

18. Switch to simulation mode (bottom right of the workspace).
19. Send a ping or web request between two devices and play/capture controls to step through the packet as it travels hop-by-hop through the switch, observing MAC-address-based forwarding.

Observation :

- All four PCs were successfully connected to the switch in a star topology and assigned unique static IP address within the same subnet (10.0.0.0/255.0.0.0). Ping tests between all devices returned successful replies, confirming end-to-end connectivity. The shared folder/server resource was successfully accessed from other computers on the network, confirming that the LAN was functioning correctly for resource sharing.

